VFSTR MAIN CAMPUS, Vadlamudi	Formative Assessment Answer Sheet	Regd. No.										
Assessment name:	Date:	Course Name:	Subject Code:	Marks Awarded								
Year: Semester:	Program: Branch:	Section:	Signature of the invigilator:									

pre - M2T1 - slot 2 topics/concepts

- Position of Fermi level in extrinsic (n and p-type) semiconductor
- Position of Fermi level at different temperature
- Circuit diagram and depletion region width under different bias conditions
- Solar cell
  - construction
  - working principle
  - I-V characteristics
    - Short circuit current
    - Open circuit voltage
  - P-V characteristics
    - Fill factor
    - Efficiency

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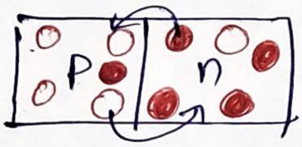
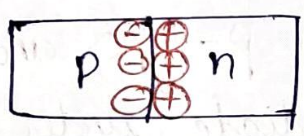
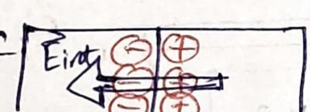
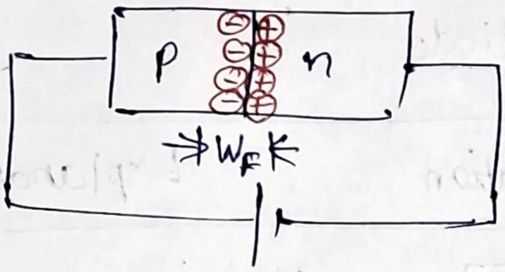
Explain formation of depletion region and built-in potential in p-n junction diode

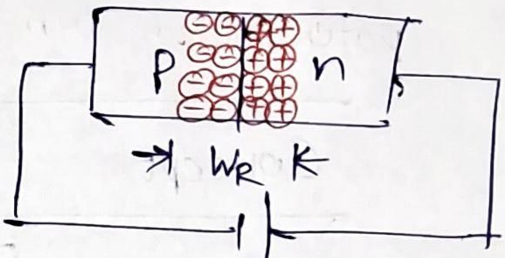
Concept	Illustration	Explanation
1. Diffusion		Majority carriers of one region diffuse into other region
2. Space charge region		Diffusion of mobile charge carriers leads to unbalanced space charges near the junction. This is called space charge region or depletion region.
3. Built-in electric field		Space charge region forms a parallel plate capacitor with electric field from n-region to p-region. This is called built-in electric field.
4. Built-in Potential	$E = \frac{V}{\cancel{d}W} \Rightarrow V = E\cancel{d}W$	If the width of depletion region is W, then the built-in potential is given by V = EW

How depletion region width varies under forward and reverse bias with neat diagrams?

Forward bias	Reverse bias



$V_{ext} = V_F$



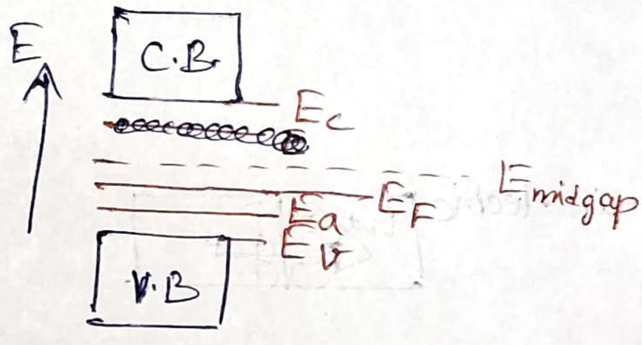
$V_{ext} = V_R$

$W_F < W_R$

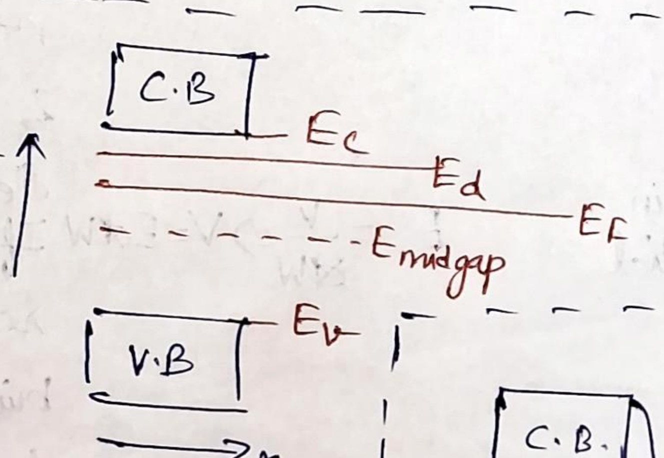
Draw Fermi level in p-type and n-type semiconductors and in p-n junction diode under equilibrium

Material	Energy band diagram (*)

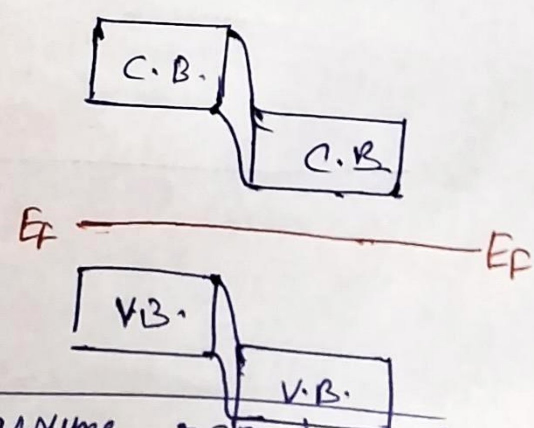
p



n



p | n



(*) If temperature is not given, we assume room temperature

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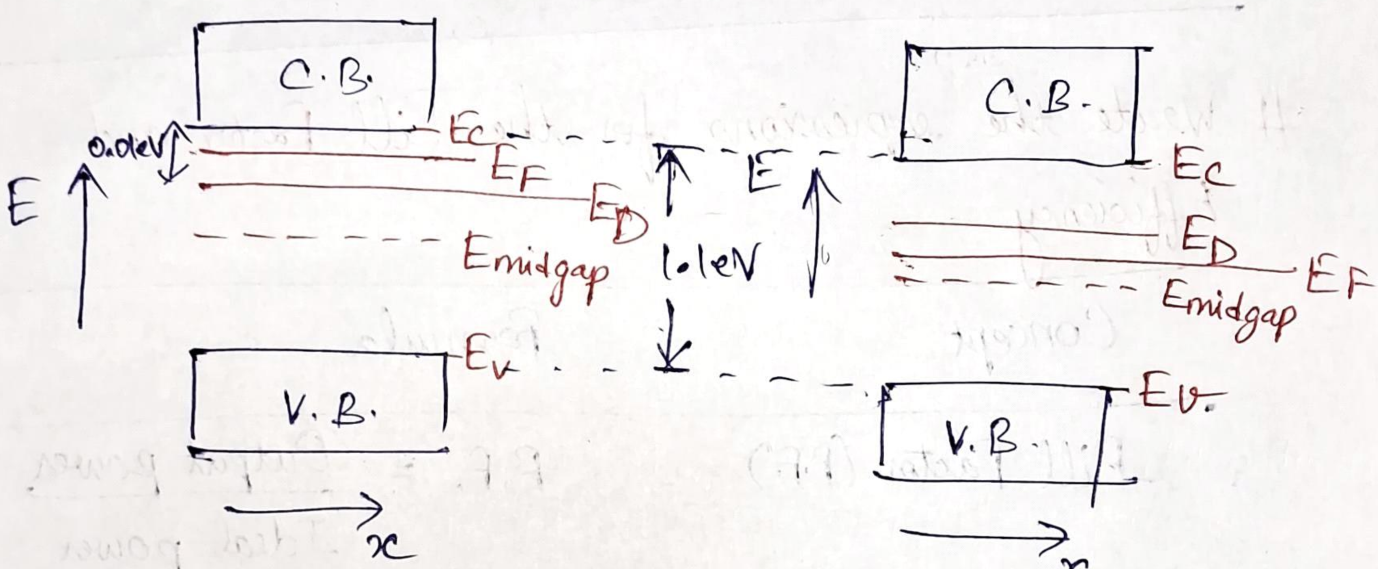
\# Assume that Si s.c. with band gap of 1.1eV is doped with P impurity. Draw band diagram of n-type s.c. @ T=0K and T=300K. Assume donor level lies below conduction of Si by 0.01eV.

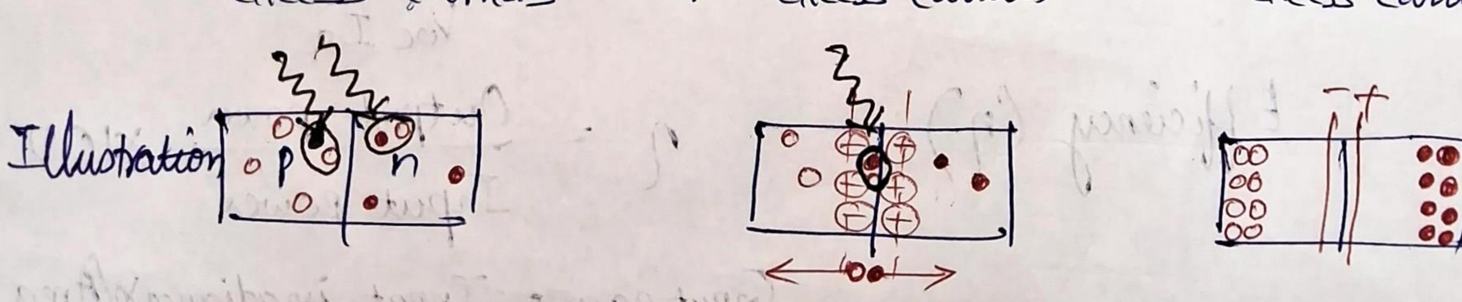
T=0K T=300K



\# Explain the role of depletion region in p-n junction diode in conversion of solar energy into electrical energy.

Photo-voltaic effect

Concept: Generation of excess carriers ⇒ Seperation of excess carriers ⇒ Accumulation of excess carriers



Explanation	Light breaks covalent bond and creates excess charge carriers Photon + covalent bond $\downarrow$ electron + hole	@ Depletion region, Built-in electric field separates the charge carriers. holes are pushed in p-region, electrons are pushed in n-region	Accumulation of separated carriers leads to generation of voltage. Accumulated holes give positive voltage, electrons give negative voltage. This is called photo-voltage

Write the expressions for the Fill Factor and Efficiency

Concept	Formula
Fill Factor (F.F.)	$F.F. = \dfrac{\text{Output power}}{\text{Ideal power}}$ Output power = (Output voltage) × (Output current) $= V_m \cdot I_m$ Ideal power = (Open circuit voltage) × (Short circuit current) $= V_{oc} I_{sc}$ $F.F. = \dfrac{V_m I_m}{V_{oc} I_{sc}}$
Efficiency (η)	$\eta = \dfrac{\text{Output power}}{\text{Input power}} \times 100$ Input power = Input irradiance × (Area of panel)